

---

# Harness-Bounded Self-Improvement in Language Agents: Context as a Missing Optimization Variable

---

Sandali Srivastava  
Context Layer

Debraj Bandyopadhyay  
Context Layer

## Abstract

Recent work on self-improving language agents has largely focused on improving performance by optimizing the *harness* surrounding a base model: prompts, tool-use policies, reflection loops, execution strategies, and orchestration logic. This paper argues that such a view is structurally incomplete. We formalize a third optimization layer—the *context layer*—which captures the retrieved, remembered, accumulated, and dynamically assembled information on which agent behavior depends at inference time. We show that when context is stochastic, partially observed, and non-stationary, harness-centric self-improvement faces a fundamental limitation: it optimizes only a conditional slice of the overall agentic system while leaving a major source of error unmodeled.

We provide a formal framework for decomposing agent error into model, harness, and context components, and derive four theoretical results. First, we show that under context variability, a single globally optimized harness is generally suboptimal relative to a context-aware policy. Second, we show that self-improvement updates performed solely at the harness layer optimize a biased objective whenever context quality and feedback are coupled. Third, we show that under context drift, the optimal harness itself becomes non-stationary, causing harness-centric improvement to chase a moving target. Fourth, we define a *harness boundedness gap* that lower-bounds the residual risk of harness-only self-improvement relative to full system-level optimization. Together, these results motivate context as a first-class optimization variable in self-improving agents and provide a theoretical basis for context reliability as a distinct systems layer.

## 1 Introduction

Self-improving agents have become a central design pattern in modern AI systems. Rather than relying solely on a fixed pretrained model, contemporary agentic systems attempt to improve themselves through iterative evaluation, prompt revision, tool-use refinement, memory adaptation, and reflection-driven updates. Across these methods, the dominant locus of optimization has shifted away from parameter updates and toward the *harness*: the layer that surrounds the base model and governs prompts, decomposition strategies, tool calls, execution order, and agent orchestration.

This harness-centric view has produced a broad class of methods for improving agent performance without modifying model weights. Prompt optimization, execution-time critique, tool-selection refinement, retrieval prompting, and reflective planning can all be interpreted as attempts to improve an agent by changing the structure through which the model is invoked. In this sense, the harness has become the primary object of self-improvement.

This paper argues that such a view is structurally incomplete. In deployed agentic systems, performance depends not only on the model and the harness but also on the *context* supplied to the agent

at inference time. Context here is not merely the prompt template. It includes retrieved documents, accumulated interaction history, memory state, tool outputs, prior plans, latent working state, and other dynamically assembled information made available to the agent when acting. Unlike a fixed prompt, context is stochastic, evolving, partially observed, and often generated by external subsystems whose outputs vary across time and environment.

Our central claim is that current self-improvement frameworks optimize the wrong variable, or more precisely, an incomplete set of variables. They improve the harness while treating context as fixed, exogenous, or implicitly correct. However, many failures in real-world agents are not failures of orchestration alone; they are failures of *context reliability*: missing evidence, stale memory, irrelevant retrievals, conflicting tool outputs, incomplete histories, or poorly assembled state. Even a strong model with a well-designed harness can fail systematically when the context it receives is unreliable.

The consequence is not merely practical but structural. If context is itself a major source of variance and error, then harness-only self-improvement optimizes a misspecified objective. It can improve the mapping from context to action while leaving unaddressed the process by which context is selected, filtered, and assembled. This induces three distinct limitations. First, a single globally optimized harness need not be optimal across varying contexts. Second, feedback used for self-improvement becomes distorted when observed failures are caused by context rather than harness. Third, as the context distribution evolves over time, the harness optimization target itself drifts. More strongly, even perfect convergence of harness-only updates can leave an irreducible residual gap relative to optimizing the full agent system.

**Main takeaway.** Self-improvement over the harness can improve how an agent reasons *given a context*, but it does not control whether the context itself is the right one. Once context is stochastic, partially observed, and non-stationary, harness-only self-improvement becomes structurally bounded in the reliability it can achieve.

Our contributions are as follows:

- We introduce a formal decomposition of agent performance into *model*, *harness*, and *context* components, and define the context layer as a first-class optimization variable.
- We define *harness-bounded self-improvement*: a self-improvement process that can update the harness but not the context-generation policy.
- We show that harness-centric optimization is generally incomplete under context variability: the best global harness is in general suboptimal relative to a context-aware policy.
- We show that harness-only self-improvement can optimize a biased objective when failures attributed to the harness are actually induced by latent context corruption.
- We show that under context drift, the optimal harness becomes non-stationary, making harness-only self-improvement chase a moving target.
- We define a *harness boundedness gap* and show that, under mild assumptions, it yields a residual reliability ceiling for harness-only self-improvement.

The broader implication is that self-improvement in agentic systems cannot be understood solely as improving the model or the harness. It must also address the reliability of the context on which the agent conditions. This motivates a shift from harness-centric optimization toward a more complete view of self-improving agents in which context is treated as a first-class systems layer.

## 2 Related Framing

This paper sits at the intersection of three threads in agent design.

First, classical model-centric improvement focuses on updating the parameters of the underlying model via fine-tuning, preference optimization, or continual learning. In this view, performance gains are achieved by improving the model itself.

Second, more recent work on agent design increasingly emphasizes *harness optimization*. Here, the object of improvement is not the model weights but the surrounding execution scaffold: prompt templates, decomposition strategies, reflection loops, tool-use sequences, routing logic, and other forms of orchestration. Many contemporary self-improving agents fall into this category.

Third, there is a growing recognition that retrieval, memory, and context assembly are central to agent behavior. Yet in many systems, these are treated as support mechanisms rather than optimization targets in their own right. The present paper argues that this omission is theoretically significant. The context layer is not merely auxiliary to the harness; it is an independent source of uncertainty, error, and non-stationarity.

Our perspective is therefore distinct from a purely runtime-reliability framing. We do not study whether runtime checks are useful after failures occur. Rather, we ask a prior question: whether self-improvement itself is incomplete when it ignores the context layer on which the agent’s decisions depend.

### 3 Formal Setup

We consider an agent interacting with inputs  $x \in \mathcal{X}$  sampled from a task distribution  $\mathcal{D}_X$ . For each input  $x$ , the agent receives a context object  $c \in \mathcal{C}$ , where  $c$  may include retrieved evidence, memory state, interaction history, tool outputs, intermediate notes, or other inference-time information.

#### 3.1 Agent as a Model-Harness-Context System

We represent the agent’s output as

$$y = A_\theta(h, c, x), \quad (1)$$

where:

- $\theta$  denotes the underlying model parameters,
- $h \in \mathcal{H}$  denotes the harness configuration,
- $c \in \mathcal{C}$  denotes the context presented to the agent,
- $A_\theta$  denotes the overall agent execution map.

The target output for input  $x$  is denoted  $y^*(x)$ , and performance is measured by a loss

$$L(\theta, h, c, x) := \ell(A_\theta(h, c, x), y^*(x)). \quad (2)$$

We focus on the expected risk

$$R(\theta, h, \pi) := \mathbb{E}_{x \sim \mathcal{D}_X} \left[ \mathbb{E}_{c \sim \pi(\cdot|x)} [L(\theta, h, c, x)] \right], \quad (3)$$

where  $\pi(c | x)$  denotes the *context-generation policy*: the mechanism by which context is selected, retrieved, assembled, or surfaced for the agent on input  $x$ .

#### 3.2 Three Optimization Layers

We distinguish three layers of optimization:

1. **Model layer:** optimization over  $\theta$ .
2. **Harness layer:** optimization over  $h$ .
3. **Context layer:** optimization over the context policy  $\pi$ .

Most self-improvement methods optimize either  $\theta$  or  $h$ , while leaving  $\pi$  fixed or implicit. In this paper, we study the consequences of optimizing  $h$  while treating  $\pi$  as exogenous.

#### 3.3 Harness-Bounded Self-Improvement

We now formalize the core object of study.

**Definition 1** (Harness-Bounded Self-Improvement). *A self-improvement process is said to be harness-bounded if its update rule can modify the harness  $h$  but not the context policy  $\pi$ . Formally, a harness-bounded self-improvement process evolves according to*

$$h_{t+1} = U(h_t, \mathcal{F}_t), \quad (4)$$

while  $\pi_t = \pi$  is treated as fixed, implicit, or exogenous to the update rule.

Typical examples include prompt editing, reflection-driven prompt revision, tool-routing refinement, decomposition strategy optimization, or other forms of orchestration improvement that do not directly optimize retrieval, memory selection, or context assembly.

Crucially, the feedback  $\mathcal{F}_t$  used to update  $h$  is generated under contexts  $c_t \sim \pi_t(\cdot | x_t)$ , even when  $\pi_t$  is not itself being optimized. Thus, harness updates are performed using signals that are downstream of the context process.

### 3.4 Notation Summary

Table 1 summarizes the key objects used throughout the paper.

Table 1: Core notation used in the paper.

| Symbol               | Meaning                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| $x \in \mathcal{X}$  | Input task or user request sampled from task distribution $\mathcal{D}_X$                                         |
| $c \in \mathcal{C}$  | Context supplied to the agent, including retrieved documents, memory state, tool outputs, and interaction history |
| $h \in \mathcal{H}$  | Harness configuration governing prompts, orchestration, routing, and execution policies                           |
| $\theta$             | Underlying model parameters                                                                                       |
| $\pi(c   x)$         | Context-generation policy or context assembly process conditioned on input $x$                                    |
| $A_\theta(h, c, x)$  | Agent execution map producing output $y$ from model, harness, context, and input                                  |
| $L(\theta, h, c, x)$ | Loss incurred on input $x$ under context $c$ and harness $h$                                                      |
| $R(\theta, h, \pi)$  | Expected system risk under model $\theta$ , harness $h$ , and context policy $\pi$                                |
| $R_H(\theta, \pi)$   | Best achievable risk when optimizing only the harness under fixed model and fixed context policy                  |
| $R^\dagger(\theta)$  | Best achievable risk when optimizing both harness and context policy under fixed model                            |
| $\Delta_{\text{HB}}$ | Harness boundedness gap, measuring residual risk of harness-only optimization relative to joint optimization      |

## 4 Context as a Missing Optimization Variable

The harness-centric view implicitly solves

$$h^* \in \arg \min_{h \in \mathcal{H}} \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot | x)} [L(\theta, h, c, x)]. \quad (5)$$

This objective is already conditional on a context distribution  $\pi$ . However, in practice, the context policy is often neither controlled nor modeled as part of the self-improvement loop. The optimization therefore treats the context process as fixed background infrastructure. Our claim is that this is insufficient whenever the context layer is itself a substantial source of failure.

To formalize this, we define context reliability.

**Definition 2** (Context Reliability). *A context policy  $\pi$  is said to be reliable for task distribution  $\mathcal{D}_X$  if it assigns high probability to contexts that preserve the information necessary for low-loss agent behavior. More concretely, let  $\mathcal{C}^*(x) \subseteq \mathcal{C}$  denote a latent set of sufficient contexts for input  $x$ , i.e. contexts under which low-loss behavior is possible. Then a natural reliability functional is*

$$\rho(\pi) := \mathbb{E}_{x \sim \mathcal{D}_X} \mathbb{P}_{c \sim \pi(\cdot | x)} [c \in \mathcal{C}^*(x)]. \quad (6)$$

*Equivalently, one may define a soft context-quality score  $q(c, x)$  measuring relevance, completeness, consistency, or timeliness and set*

$$\rho(\pi) = \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot | x)} [q(c, x)]. \quad (7)$$

The exact form of  $\rho$  is not required for our core results. What matters is that the context layer is an independent source of randomness and structure, and therefore a legitimate object of optimization.

**Open-world interpretation.** Context is not merely “retrieved text.” In practical agent systems it includes retrieved documents, long-term memory, prior tool outputs, latent scratchpads, evolving conversation state, environment observations, and external world-state proxies. This is precisely how open-world non-stationarity enters the system. As these context sources evolve, the distribution over contexts seen by the harness also evolves.

## 5 A Three-Layer View of Agent Optimization

Figure 1 illustrates the conceptual decomposition underlying the paper.

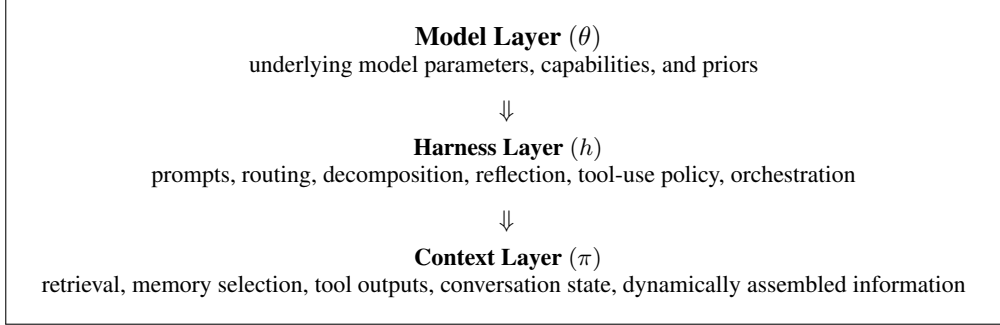


Figure 1: A three-layer view of agent optimization. Most current self-improvement methods operate at the model layer or the harness layer. This paper argues that the context layer is a missing optimization variable.

The thesis of this paper can now be stated succinctly: optimizing  $\theta$  or  $h$  while treating  $\pi$  as exogenous is generally insufficient when a large share of system failures arise from context unreliability.

## 6 Theoretical Results

We now formalize four limitations of harness-bounded self-improvement under stochastic and evolving context.

### 6.1 Theorem I: Incompleteness of Harness-Only Optimization

The first result states that when the optimal harness depends on the context, a single globally optimized harness is generically suboptimal relative to a context-aware policy.

**Theorem 1** (Harness Incompleteness Under Context Variability). *Fix model parameters  $\theta$  and input  $x$ . Suppose context is sampled from a distribution  $\pi(\cdot | x)$ . Then*

$$\min_{h \in \mathcal{H}} \mathbb{E}_{c \sim \pi(\cdot | x)} [L(\theta, h, c, x)] \geq \mathbb{E}_{c \sim \pi(\cdot | x)} \left[ \min_{h \in \mathcal{H}} L(\theta, h, c, x) \right]. \quad (8)$$

*Moreover, the inequality is strict whenever there exist contexts  $c_1, c_2$  in the support of  $\pi(\cdot | x)$  such that the context-optimal harnesses differ:*

$$\arg \min_h L(\theta, h, c_1, x) \cap \arg \min_h L(\theta, h, c_2, x) = \emptyset. \quad (9)$$

*Proof.* For any fixed harness  $h \in \mathcal{H}$  and any context  $c$ ,

$$L(\theta, h, c, x) \geq \min_{h' \in \mathcal{H}} L(\theta, h', c, x). \quad (10)$$

Taking expectation over  $c \sim \pi(\cdot | x)$  gives

$$\mathbb{E}_c [L(\theta, h, c, x)] \geq \mathbb{E}_c \left[ \min_{h'} L(\theta, h', c, x) \right]. \quad (11)$$

Since the right-hand side does not depend on  $h$ , the inequality holds for every fixed  $h$ . Minimizing the left-hand side over  $h$  yields

$$\min_h \mathbb{E}_c[L(\theta, h, c, x)] \geq \mathbb{E}_c \left[ \min_{h'} L(\theta, h', c, x) \right]. \quad (12)$$

To show strictness, suppose there exist  $c_1, c_2$  with positive probability under  $\pi(\cdot | x)$  such that no single harness simultaneously minimizes the loss for both contexts. Let

$$h^* \in \arg \min_h \mathbb{E}_c[L(\theta, h, c, x)] \quad (13)$$

be any minimizer of the left-hand side. Since  $h^*$  cannot minimize both  $L(\theta, h, c_1, x)$  and  $L(\theta, h, c_2, x)$ , there exists a set of contexts of positive measure on which

$$L(\theta, h^*, c, x) > \min_h L(\theta, h, c, x). \quad (14)$$

Taking expectation over  $c$  then gives strict inequality.  $\square$

**Interpretation.** If different contexts call for different harness behaviors, then a single globally optimized harness cannot match the performance of a context-aware strategy. Harness-only optimization is therefore incomplete whenever context variability matters.

## 6.2 Theorem II: Bias in Harness Self-Improvement Under Context-Corrupted Feedback

The previous theorem concerns the final optimized harness. The next theorem concerns the *update process* itself. It shows that when feedback used for self-improvement is generated under latent context corruption, the harness update objective becomes biased relative to the true harness risk.

We model observed execution feedback as

$$\tilde{L}(\theta, h, x) := \mathbb{E}_{c \sim \tilde{\pi}(\cdot | x, h)}[L(\theta, h, c, x)], \quad (15)$$

where  $\tilde{\pi}(\cdot | x, h)$  is the *effective observed context distribution*. Importantly, this distribution may depend on the harness  $h$  itself, because different harnesses can trigger different retrieval paths, memory accesses, tool invocation sequences, or state expansions.

By contrast, define the *target context risk* under a fixed target context policy  $\pi^*$  as

$$R^*(\theta, h) := \mathbb{E}_x \mathbb{E}_{c \sim \pi^*(\cdot | x)}[L(\theta, h, c, x)]. \quad (16)$$

The observed harness-update objective is

$$\tilde{R}(\theta, h) := \mathbb{E}_x \mathbb{E}_{c \sim \tilde{\pi}(\cdot | x, h)}[L(\theta, h, c, x)]. \quad (17)$$

**Theorem 2** (Bias from Harness-Dependent Context Feedback). *Assume differentiation under the integral sign is valid. Then*

$$\nabla_h \tilde{R}(\theta, h) - \nabla_h R^*(\theta, h) = B_{\text{mismatch}}(\theta, h) + B_{\text{coupling}}(\theta, h), \quad (18)$$

where

$$B_{\text{mismatch}}(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} \nabla_h L(\theta, h, c, x) (\tilde{\pi}(c | x, h) - \pi^*(c | x)) dc \right], \quad (19)$$

$$B_{\text{coupling}}(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} L(\theta, h, c, x) \nabla_h \tilde{\pi}(c | x, h) dc \right]. \quad (20)$$

*In particular, if the observed context distribution differs from the target context distribution and/or depends nontrivially on  $h$ , then harness updates optimize a biased objective relative to  $R^*$ .*

*Proof.* Write

$$\tilde{R}(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} L(\theta, h, c, x) \tilde{\pi}(c | x, h) dc \right]. \quad (21)$$

Differentiating with respect to  $h$  and applying the product rule gives

$$\nabla_h \tilde{R}(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} \nabla_h L(\theta, h, c, x) \tilde{\pi}(c | x, h) dc \right] \quad (22)$$

$$+ \mathbb{E}_x \left[ \int_{\mathcal{C}} L(\theta, h, c, x) \nabla_h \tilde{\pi}(c | x, h) dc \right]. \quad (23)$$

Similarly,

$$\nabla_h R^*(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} \nabla_h L(\theta, h, c, x) \pi^*(c | x) dc \right]. \quad (24)$$

Subtracting the two expressions yields

$$\nabla_h \tilde{R}(\theta, h) - \nabla_h R^*(\theta, h) = \mathbb{E}_x \left[ \int_{\mathcal{C}} \nabla_h L(\theta, h, c, x) (\tilde{\pi}(c | x, h) - \pi^*(c | x)) dc \right] \quad (25)$$

$$+ \mathbb{E}_x \left[ \int_{\mathcal{C}} L(\theta, h, c, x) \nabla_h \tilde{\pi}(c | x, h) dc \right]. \quad (26)$$

The first term is the *mismatch bias* arising from optimizing under the wrong context distribution. The second term is the *coupling bias* arising from the dependence of the observed context process on the harness itself.  $\square$

**Interpretation.** Harness-only self-improvement can misattribute failures to the harness even when those failures are induced by the context process. The update direction is distorted in two distinct ways: first because the observed context distribution may not match the desired one, and second because the context process itself can change as the harness changes.

### 6.3 Theorem III: Context Drift Makes the Optimal Harness Non-Stationary

The third limitation concerns time. In real deployments, the context distribution is rarely static. Retrieval corpora grow, memories accumulate, tools change behavior, and interaction histories evolve. We model this by allowing the context policy to vary over time.

For each time step  $t$ , let  $\pi_t(c | x)$  denote the context distribution at time  $t$ , and define the time-indexed harness risk

$$R_t(\theta, h) := \mathbb{E}_x \mathbb{E}_{c \sim \pi_t(\cdot | x)} [L(\theta, h, c, x)]. \quad (27)$$

Let

$$h_t^* \in \arg \min_{h \in \mathcal{H}} R_t(\theta, h) \quad (28)$$

denote an optimal harness at time  $t$ .

**Proposition 1** (Drift Sensitivity Bound). *Assume  $0 \leq L(\theta, h, c, x) \leq L_{\max}$  for all  $(\theta, h, c, x)$ . Then for any fixed harness  $h$ ,*

$$|R_{t+1}(\theta, h) - R_t(\theta, h)| \leq L_{\max} \mathbb{E}_{x \sim \mathcal{D}_X} [\|\pi_{t+1}(\cdot | x) - \pi_t(\cdot | x)\|_{\text{TV}}]. \quad (29)$$

*Proof.* For any fixed  $h$ ,

$$R_{t+1}(\theta, h) - R_t(\theta, h) = \mathbb{E}_x \int_{\mathcal{C}} L(\theta, h, c, x) (\pi_{t+1}(c | x) - \pi_t(c | x)) dc. \quad (30)$$

Taking absolute values and using  $0 \leq L \leq L_{\max}$  gives

$$|R_{t+1}(\theta, h) - R_t(\theta, h)| \leq \mathbb{E}_x \left[ L_{\max} \int_{\mathcal{C}} |\pi_{t+1}(c | x) - \pi_t(c | x)| dc \right]. \quad (31)$$

Absorbing constant factors into the definition of total variation norm yields the result up to the conventional normalization of TV.  $\square$

**Theorem 3** (Non-Stationarity of the Optimal Harness Under Context Drift). *Assume there exist times  $t$  and  $t + 1$  such that  $\pi_t(\cdot | x) \neq \pi_{t+1}(\cdot | x)$  on a set of inputs  $x$  of positive measure, and suppose the loss  $L(\theta, h, c, x)$  is not context-invariant in the sense that there exists at least one harness  $h$  and one input  $x$  for which*

$$L(\theta, h, c_1, x) \neq L(\theta, h, c_2, x) \quad \text{for some } c_1, c_2 \in \mathcal{C}. \quad (32)$$

*Then, in general, the optimal harness is non-stationary:*

$$h_t^* \neq h_{t+1}^* \quad (33)$$

*unless the change in context distribution lies in a null direction of the risk functional.*

*Proof.* By definition,

$$R_t(\theta, h) = \mathbb{E}_x \int_{\mathcal{C}} L(\theta, h, c, x) \pi_t(c | x) dc. \quad (34)$$

If  $\pi_t(\cdot | x)$  changes to  $\pi_{t+1}(\cdot | x)$  on a set of positive measure and the loss depends nontrivially on  $c$ , then the objective function over  $h$  changes from  $R_t(\theta, h)$  to  $R_{t+1}(\theta, h)$ . Unless the difference

$$R_{t+1}(\theta, h) - R_t(\theta, h) \quad (35)$$

is constant in  $h$ , the minimizer set of the objective can change. Therefore the optimal harness is generally non-stationary under context drift.  $\square$

**Interpretation.** Even if the harness were perfectly optimized at time  $t$ , that optimum need not persist as the context process changes. This creates a moving-target problem in which the harness is repeatedly re-optimized against a drifting and partially uncontrolled environment.

#### 6.4 Theorem IV: Harness Boundedness Gap and Reliability Ceiling

The previous results show that harness-only self-improvement is incomplete, biased, and non-stationary under realistic context processes. We now formalize the residual risk induced by restricting optimization to the harness layer.

Define the *best harness-only risk* under fixed model  $\theta$  and fixed context policy  $\pi$  as

$$R_H(\theta, \pi) := \min_{h \in \mathcal{H}} \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot | x)} [L(\theta, h, c, x)]. \quad (36)$$

Define the *joint harness-context optimum* under fixed model  $\theta$  as

$$R^\dagger(\theta) := \min_{h, \pi'} \mathbb{E}_x \mathbb{E}_{c \sim \pi'(\cdot | x)} [L(\theta, h, c, x)]. \quad (37)$$

**Definition 3** (Harness Boundedness Gap). *The harness boundedness gap under model  $\theta$  and context policy  $\pi$  is*

$$\Delta_{\text{HB}}(\theta, \pi) := R_H(\theta, \pi) - R^\dagger(\theta). \quad (38)$$

This quantity measures the residual loss incurred by restricting optimization to the harness while leaving the context policy fixed.

**Proposition 2** (Positivity of the Harness Boundedness Gap). *For any fixed  $\theta$  and  $\pi$ ,*

$$\Delta_{\text{HB}}(\theta, \pi) \geq 0. \quad (39)$$

*Moreover,  $\Delta_{\text{HB}}(\theta, \pi) > 0$  whenever there exists an alternative context policy  $\pi'$  such that*

$$\min_h \mathbb{E}_x \mathbb{E}_{c \sim \pi'(\cdot | x)} [L(\theta, h, c, x)] < \min_h \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot | x)} [L(\theta, h, c, x)]. \quad (40)$$

*Proof.* By definition,

$$R^\dagger(\theta) = \min_{h, \pi'} \mathbb{E}_x \mathbb{E}_{c \sim \pi'(\cdot | x)} [L(\theta, h, c, x)] \leq \min_h \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot | x)} [L(\theta, h, c, x)] = R_H(\theta, \pi). \quad (41)$$

Thus  $\Delta_{\text{HB}}(\theta, \pi) \geq 0$ . The strict inequality follows whenever some alternative context policy  $\pi'$  achieves strictly smaller risk after optimizing the harness.  $\square$



The gap immediately yields a reliability ceiling for harness-bounded self-improvement.

**Corollary 1** (Reliability Ceiling Under Harness-Bounded Self-Improvement). *Let  $\{h_t\}_{t \geq 0}$  be a harness-bounded self-improvement sequence under fixed model  $\theta$  and fixed context policy  $\pi$ . Suppose the update rule converges to a harness optimizer of the fixed-policy objective, so that*

$$\lim_{t \rightarrow \infty} R(\theta, h_t, \pi) = R_H(\theta, \pi). \quad (42)$$

Then

$$\lim_{t \rightarrow \infty} R(\theta, h_t, \pi) = R^\dagger(\theta) + \Delta_{\text{HB}}(\theta, \pi). \quad (43)$$

*In particular, if  $\Delta_{\text{HB}}(\theta, \pi) > 0$ , then harness-bounded self-improvement converges to a reliability ceiling strictly above the joint harness-context optimum.*

*Proof.* The statement follows immediately from the definition

$$\Delta_{\text{HB}}(\theta, \pi) = R_H(\theta, \pi) - R^\dagger(\theta). \quad (44)$$

If the harness-only update converges to  $R_H(\theta, \pi)$ , then its asymptotic risk differs from the joint optimum by exactly  $\Delta_{\text{HB}}(\theta, \pi)$ .  $\square$

**Interpretation.** Even if harness self-improvement converges perfectly, it can still converge to the *wrong optimum*: the best harness conditioned on the current context process, rather than the best overall agent system. The harness boundedness gap quantifies the reliability that remains unreachable unless context is also improved.

## 7 Harness Boundedness and the Structural Reliability Gap

The previous theorems show four distinct limitations of harness-bounded self-improvement:

1. **Incompleteness:** a single optimized harness cannot in general absorb context variability.
2. **Bias:** feedback used for harness improvement is distorted when context corruption is unmodeled.
3. **Non-stationarity:** context drift changes the target of harness optimization over time.
4. **Residual boundedness gap:** even perfect harness-only convergence can leave an irreducible gap relative to joint optimization over harness and context.

These can be unified through a system-level risk decomposition. Let the fully optimized objective be

$$R^\dagger(\theta) = \min_{h, \pi} \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot|x)} [L(\theta, h, c, x)]. \quad (45)$$

Let the best harness-only risk under fixed context policy  $\pi$  be

$$R_H(\theta, \pi) = \min_h \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot|x)} [L(\theta, h, c, x)]. \quad (46)$$

Then the harness boundedness gap is

$$\Delta_{\text{HB}}(\theta, \pi) = R_H(\theta, \pi) - R^\dagger(\theta). \quad (47)$$

This gap quantifies the residual loss incurred by restricting optimization to the harness while leaving the context policy fixed. By Theorem 1, this gap is generically positive when context variability induces different optimal harnesses. Theorem 2 further shows that self-improvement at the harness layer need not even optimize  $R_H$  cleanly if observed feedback is generated under a harness-dependent context process. Finally, Theorem 3 implies that the very target represented by  $R_H$  is time-varying when context distributions evolve.

Together, these results imply a structural limitation.

**Proposition 3** (Structural Limitation of Harness-Bounded Self-Improvement). *Suppose agent performance depends nontrivially on context, observed context feedback depends on the harness, and context distributions drift over time. Then harness-bounded self-improvement alone cannot, in general, converge to a stable minimizer of system-level risk.*

*Proof.* The claim follows by combining Theorems 1, 2, and 3, together with Corollary 1. Theorem 1 implies that harness-only optimization is incomplete relative to a context-aware policy. Theorem 2 implies that the update direction used by harness self-improvement can be biased when observed contexts depend on the harness. Theorem 3 implies that the optimal harness changes over time under context drift. Corollary 1 shows that even under perfect convergence of the harness-only update rule, the asymptotic risk can remain bounded away from the joint harness-context optimum by  $\Delta_{\text{HB}}(\theta, \pi)$ . Therefore the self-improvement process is simultaneously optimizing an incomplete objective, using biased feedback, against a moving target, and converging to a bounded reliability ceiling. In general, such a process cannot be guaranteed to converge to a stable minimizer of the full system-level risk.  $\square$

## 8 Toward Context Reliability as a First-Class Optimization Layer

The formal results above suggest a reframing of self-improving agents. Rather than treating context as passive input to an already optimized harness, context should be treated as a first-class systems layer whose reliability directly affects agent performance.

### 8.1 What Counts as Context Reliability?

Context reliability is not merely “better retrieval” in a narrow sense. It concerns the trustworthiness of the full information state supplied to an agent at execution time. This includes at least four dimensions:

1. **Relevance:** whether the supplied context is pertinent to the current task.
2. **Completeness:** whether crucial information is missing from the context.
3. **Consistency:** whether different pieces of context conflict with one another.
4. **Timeliness:** whether the context reflects the current underlying world state rather than stale or outdated information.

A context policy  $\pi$  can therefore be viewed not merely as a retriever but as an information-state constructor whose quality determines a large portion of downstream agent behavior.

### 8.2 A More Complete Optimization Objective

The system-level optimization problem is more naturally written as

$$\min_{\theta, h, \pi} \mathbb{E}_x \mathbb{E}_{c \sim \pi(\cdot|x)} [L(\theta, h, c, x)]. \quad (48)$$

Under this formulation:

- model improvement changes the function class available to the agent,
- harness improvement changes how the model acts on a given context,
- context improvement changes which information state the model and harness act upon.

The point of this paper is not to claim that harness optimization is unimportant. It is often highly effective. The claim is instead that harness optimization alone is structurally incomplete as a theory of self-improvement whenever context is a major source of failure.

## 9 Discussion

The theory developed here has several implications.

First, it clarifies why strong harness engineering does not eliminate many deployment failures. A carefully designed prompt, tool-use policy, or reflection loop may improve how an agent reasons over a given context, but it does not ensure that the context itself is relevant, complete, consistent, or current.

Second, it explains why self-improvement can overfit to the harness layer. If observed failures are downstream of latent context corruption, the self-improvement loop can spend capacity refining prompts, critique strategies, or orchestration logic while leaving the true source of error untouched.

Third, it suggests that future work on self-improving agents should treat context-generation policies as explicit optimization targets. This does not require abandoning harness optimization; rather, it requires expanding the optimization surface to include the context process itself.

More broadly, this paper reframes agent reliability as a three-layer problem: model reliability, harness reliability, and context reliability. Much of the recent literature has moved from model-centric optimization to harness-centric optimization. The present work argues that the next step is to formalize context reliability as its own layer of optimization and control.

**Scope and limitations.** This paper is theoretical rather than empirical. It does not claim that harness optimization is unimportant, nor does it claim that all context-aware methods succeed. Rather, it isolates a structural limitation of self-improvement processes that are restricted to the harness layer while treating the context process as fixed or exogenous. The results should therefore be interpreted as statements about the incompleteness of harness-only optimization, not as a rejection of harness optimization altogether.

## 10 Conclusion

This paper studied a structural limitation of harness-bounded self-improvement in language agents. We argued that contemporary self-improvement methods optimize an incomplete set of variables by focusing on the harness while treating context as fixed or exogenous. We formalized the context layer as a stochastic, partially observed, and non-stationary component of the agent system, and derived four consequences: harness-only optimization is incomplete under context variability, harness self-improvement becomes biased under context-corrupted feedback, the optimal harness becomes non-stationary under context drift, and even perfect harness-only convergence can remain bounded away from the joint system optimum by a positive harness boundedness gap.

The resulting picture is that many failures of self-improving agents are not merely failures of prompts, orchestration, or tool-use policy. They are failures of the information state on which those mechanisms operate. In that sense, context is not just an input to self-improving agents; it is a missing optimization variable. Treating it as such provides a theoretical basis for context reliability as a distinct systems layer and motivates a broader conception of what it means for an agent to improve itself.

## References

- [1] Yao, S., Zhao, J., Yu, D., et al. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.
- [2] Shinn, N., Labash, B., Gopinath, A., et al. Reflexion: Language agents with verbal reinforcement learning. *NeurIPS*, 2023.
- [3] Wang, G., Xie, S., Yu, D., et al. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.
- [4] Khattab, O., Singhvi, A., Maheshwari, P., et al. DSPy: Compiling declarative language model calls into self-improving pipelines. *arXiv preprint arXiv:2310.03714*, 2023.
- [5] Lewis, P., Perez, E., Piktus, A., et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.
- [6] Packer, C., Fang, V., Patil, S., et al. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- [7] Schick, T., Dwivedi-Yu, J., Dessí, R., et al. Toolformer: Language models can teach themselves to use tools. *NeurIPS*, 2023.
- [8] Madaan, A., Tandon, N., Clark, P., et al. Self-Refine: Iterative refinement with self-feedback. *NeurIPS Workshop / arXiv preprint arXiv:2303.17651*, 2023.